

RESEARCH PROFILE

Researcher at the intersection of machine learning, mathematical optimization, and finance, focusing on tabular data and time-series modeling. Current work combines evolutionary computation and Transformer architectures to improve predictive performance, efficiency, and interpretability.

Areas: Tabular machine learning · Time series · Evolutionary computation · Transformers · Financial AI · Interpretable ML

EDUCATION

Ph.D., Computer Science & Engineering	Exp. 2027
Seoul National University · Optimization Lab	
M.S., Computer Science & Engineering	2024
Seoul National University · Optimization Lab	
B.S., Mathematics	2018
New York University	

RESEARCH & APPLIED EXPERIENCE

Agency for Defense Development	2026
Joint Research Project	
• Contributed to research on interpretable modeling and transparent model behavior.	
PFCT	2025
Joint Research Project	
• Developed a default-prediction model using tabular Transformers for financial data.	
Think Pool	2022–2024
Joint Research Project	
• Developed AI-driven financial factors for quantitative finance.	
Republic of Korea Air Force	2019–2022
Interpretation Officer	

TEACHING EXPERIENCE

Head TA · Data Structures	2023 · 2024 · 2025
Seoul National University	
Head TA · Algorithms	2023 · 2025 · 2026
Seoul National University	

Prepared course content, coordinated teaching support, answered questions, and graded assignments and examinations.

PUBLICATIONS

1. RTSC: Return-Aware Token-Space Contrastive Transformer for Cross-Sectional Stock Return Prediction

Hanbin Koo^{*}, Han Joon Byun^{*}, and Byung-ro Moon

^{*} Equal contribution.

Submitted to ICAIF 2026 · Under review
2. Label-Noise Decay Confounds Loss-Based Drift Attribution: A False-Positive Channel and a Noise-Aware Repair

Huh Joon and Han Joon Byun

Submitted to Transactions on Machine Learning Research (TMLR) · Under review
3. Multi-Objective Evolution of Patch Tokens for Tabular Transformers

Han Joon Byun, Joon Huh, Jieun Yook, and Byung-ro Moon

Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '26), 2026

doi.org/10.1145/3795095.3805076
4. Evolutionary Feature Construction for Tabular Transformers via Gradient-Enhanced Attention Rollout

Han Joon Byun, Jieun Yook, Hanbin Koo, and Byung-ro Moon

Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '26 Companion), 2026

doi.org/10.1145/3795101.3805282
5. CycleFormer: TSP Solver Based on Language Modeling

Jieun Yook, Junpyo Seo, Joon Huh, Han Joon Byun, and Byung-ro Moon

arXiv preprint, 2024

arxiv.org/abs/2405.20042
6. Evolutionary Pruning of Deep Convolutional Networks by a Memetic GA with Sped-Up Local Optimization and GLCM Energy Z-Score

Ha Hyun Cho, Han Joon Byun, Min Kee Kim, Joon Huh, and Byung-Ro Moon

Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '23 Companion), 2023

doi.org/10.1145/3583133.3590604

RESEARCH TRAJECTORY

Neural Network Foundations
CNN and ANN pruning and transfer learning.
Combinatorial Optimization
Transformer-based language modeling for the Traveling Salesperson Problem.
Financial & Tabular AI
Optimization and neural methods for performance, efficiency, and interpretability.